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Digital-twin-enabled JIT design of rock tunnel: Methodology and application

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ABSTRACT

An accurate and timely design of individual processes in drill and blast tunneling is essential for robust and low-risk construction of rock tunnels. Conventional dynamic design method of rock tunnel focuses on the modification of preliminary design schemes using exposed on-site data. A novel conceptual model, just-in-time (JIT) design of rock tunnel, is proposed in this study to denote the intrinsic requirement that tunnel support parameters should be determined in a JIT way when the rock mass are excavated. Specifically, implementation of JIT design of rock tunnel can be divided into three steps, i.e., acquisition of on-site raw data, interpretation of on-site raw data, and aggregation of the interpreted results to obtain a revised design scheme. Time consumption, accuracy and degree of automation are selected as three main indicators to measure the performance of such conceptual model. Moreover, to further improve the efficiency and accuracy, digital twin (DT) is integrated into the implementation of JIT design of rock tunnel, where Scene-Entity-Relationship-Incident-Control (SERIC) modelling method is proposed as a new DT application mode for share and reuse across the domain. The proposed DT-enabled JIT design of rock tunnel is applied into a practical engineering project, where several advanced data acquisition, interpretation and aggregation approaches are integrated using DT. Results show that the design efficiency and accuracy have been largely improved and the proposed method could facilitate information sharing between end-users and stakeholders.

1. Introduction

An accurate design of individual processes in drill and blast (D&B) tunneling is essential for robust and low-risk construction of rock tunnels. Due to the uncertainty and variability of geological conditions, field engineers need to revise the preliminary design of the tunnel using the newly-exposed information, and the two-stage design process is also called the dynamic design of the rock tunnel (Feng et al., 2016). Hence, in D&B tunneling, the efficiency and accuracy of the obtained newly-exposed on-site information greatly influence the dynamic design of the rock tunnel.

The key to successful implementation of dynamic design is quick and accurate acquisition of geological information and structural response during the construction, as well as timely feedback of design changes to construction. Conventionally, field engineers tend to use equipment such as geological compass and roughness profile gauge to record on-site information (Reid and Harrison, 2000). However, such practice is plagued with the problem of being error-prone and laborious as the efficiency of information acquisition mainly lies on the engineer experience. With the advancement of information technology (IT), multiple research efforts have been allocated to apply IT to individual processes in rock tunnel design to improve efficiency and accuracy. For instance, techniques such as digital photogrammetry (Li et al., 2019; Zhang et al., 2020), terrestrial laser scanning (Cacciari and Futai, 2016; Li et al., 2016) and machine learning (Liu et al., 2020; Wang et al., 2021) have all been implemented into rock tunnel design. Though these implementations have improved dynamic optimization efficiency of the design work to a certain level, the disconnection between individual processes still leads to poor coupling between design and construction as the existing studies mainly focus on one individual process (e.g., data acquisition, interpretation or analysis) in the design. From a holistic design and construction perspective, a novel conceptual model, just-in-time (JIT) design of rock tunnel, is proposed in this study to denote the intrinsic requirement of the rock tunnel support that support parameters need to be revised and determined when the rock mass are exposed during construction in a JIT way. JIT design of rock tunnel aims to use advanced

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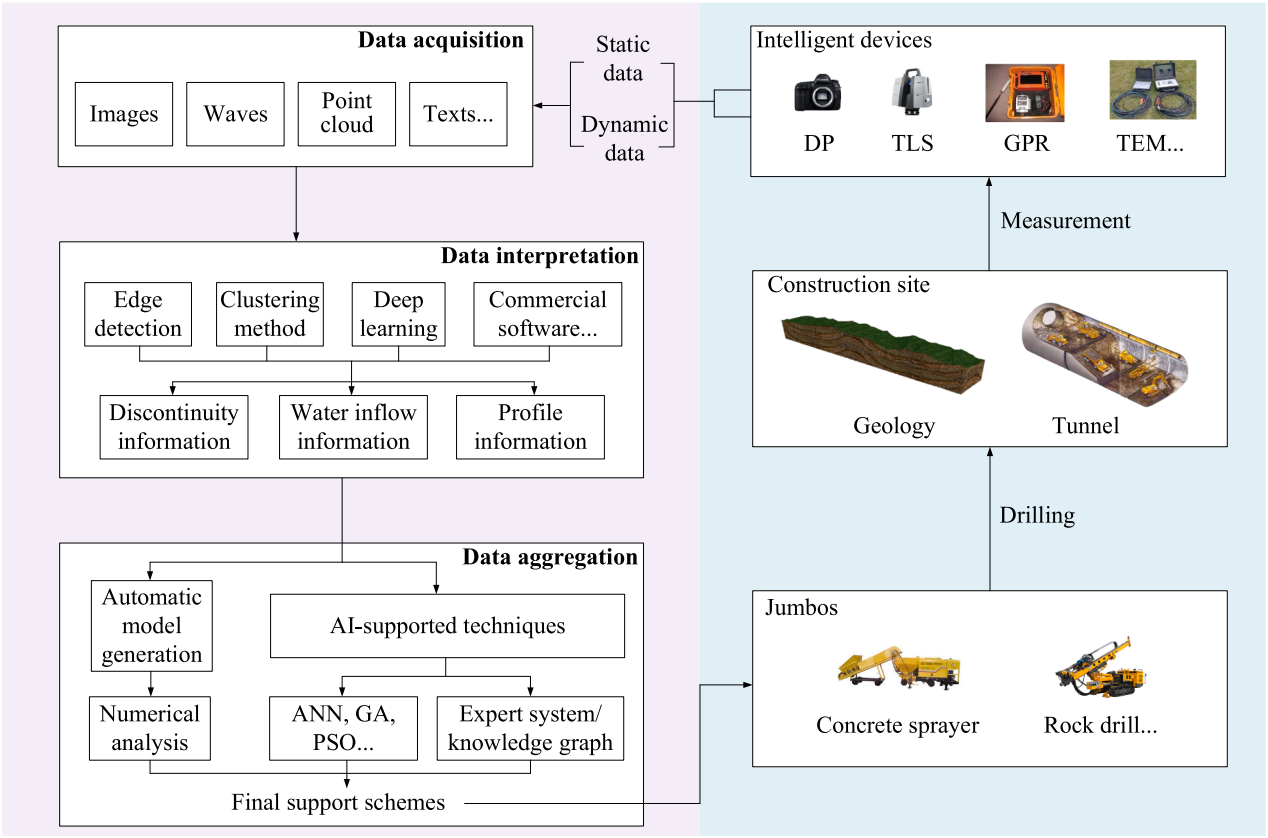


Fig. 1. Overview of JIT design of rock tunnel support.

IT technologies to assist the acquisition-interpretation-aggregation workflow and incorporate the design work into construction to form close integration. However, such conceptual framework has not been implemented in the practical engineering project. Hence, the first research question this study aims to answer is how to conduct the JIT design of rock tunnel using existing IT technologies.

The key for JIT design of rock tunnel is to improve efficiency. Recently, DT has been increasingly used in the architecture, engineering and construction (AEC) industry for its anticipated benefits such as improved productivity and efficiency (Bosch-Sijtsema et al., 2021). DT is the connection loop between physical system and virtual system which can synchronize for monitoring, simulating and optimizing purposes in a timely manner (Almatared et al., 2022; Boje et al., 2020). Actually, the core of DT is consistent with the goal of JIT design, which is to improve efficiency. Hence, it comes naturally that DT paradigm can be incorporated into tunnel design to improve design efficiency and accuracy. This has been verified by Shen et al. (2022) who used virtual world simulation and real world validation to design the luminaires inside the tunnel and found DT paradigm could contribute to design work. However, existing studies rarely apply DT to the whole design and construction process to assist rock tunnel design. As for the application mode of DT, some researches have been carried out trying to propose a unified framework for DT application (Bamunuarachchi et al., 2021; Zheng et al., 2019). In AEC industry, in most cases (such as Ding et al. (2018); Ritto and Rochinha, 2021; Yuan et al., 2016), a DT-oriented framework was also firstly put forward to show how to build DT of the target domain and then the physical- and digital- twin was set up based on the framework. However, this kind of method cannot be shared or reused as the established architecture or framework was different. In the study carried out by Grieves and Vickers (2017), DT was divided into three types: the DT Prototype, the DT Instance and the DT Aggregate. Using this type of modeling method, McClellan et al. (2022) described the construction of a two-level DT Instance for autonomous unmanned aerial vehicle landing. However, the modeling method of DT is too general to be used in rock tunnel design. Actually, a detailed and structured modeling method of DT can have important implications to not only rock tunnel design but also other domains in AEC industry as it shapes a reusable and easy-to-share paradigm for DT application. Motivated by such need, the second research question this study aims to investigate is how to build the DT of the rock tunnel during construction in an easy-to-share and –reuse way to better conduct the JIT design work.

Focusing on the two research questions, this study proposes SERIC modeling method for DT application and applies the proposed SERIC modeling method to a practical rock tunnel project, where DT is integrated into the JIT design of the tunnel. The remaining sections of this paper are organized as follows: Section 2 introduces JIT design of rock tunnel, as well as SERIC modeling method of DT and explains how to integrate DT into JIT design of rock tunnel. Section 3 is a case study which implements the proposed framework based on a rock tunnel under construction in Yunnan Province, China. Finally, Section 4 and 5 present limitations and conclusions.

2. Conceptual framework development: JIT design of rock tunnel

In this section, the above-mentioned conceptual framework of JIT design of rock tunnel is discussed in detail, including framework overview, relationships with widely-known dynamic design of rock tunnel and how DT can be incorporated into JIT design of rock tunnel to further increase design efficiency.

2.1. Framework overview

According to previous study on tunnel support design, time consumption and accuracy, as well as degree of automation, are main

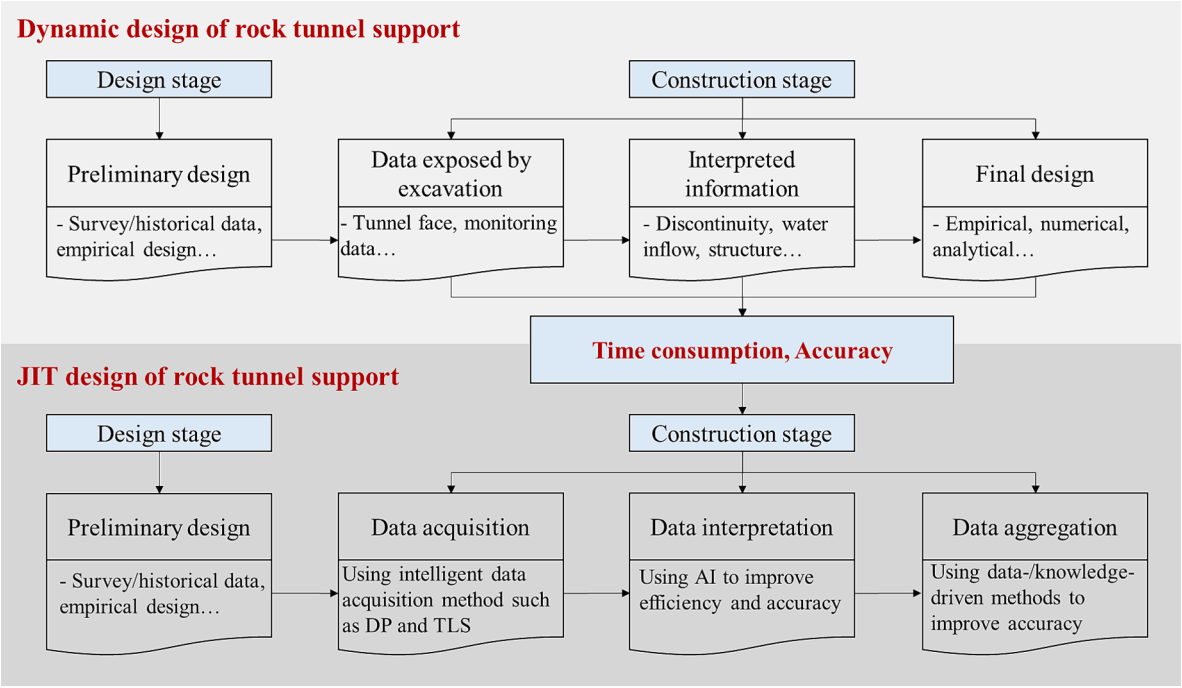


Fig. 2. Comparison between JIT design of rock tunnel support and dynamic design of rock tunnel support.

factors that influence the performance of tunnel support design (Ling et al., 2022). Hence, the starting point of JIT design of rock tunnel is to reduce time consumption and improve design accuracy of tunnel support. The term "just-in-time" denotes that the support parameters need to be determined as soon as the hidden volume of the rock mass is exposed during tunnel construction.

As mentioned above, the design of rock tunnel support can be divided into three steps, i.e. data acquisition, interpretation and aggregation. Data acquisition refers to acquisition of raw data on site such as images of tunnel face and waves of geological prospecting to record newly-exposed geological conditions, data interpretation refers to processing of on-site raw data to extract useful information for support design, and data aggregation means using processed data to give the revised support schemes. Previous studies have shown that many different technologies have been applied in the design of rock tunnel support. Considering that different technologies in the data "acquisition-interpretation-aggregation" workflow have pros and cons in terms of time consumption and accuracy, in the JIT design of rock tunnel, technologies with acceptable time consumption and accuracy are adopted.

Fig. 1 presents the overall workflow of JIT design of rock tunnel. Technologies such as digital photogrammetry (DP) (Garcia-Luna et al., 2019), terrestrial laser scanning (TLS) (Monsalve et al., 2019), ground penetrating radar (GPR) and transient electromagnetic method (TEM) (Ling et al., 2022) are incorporated in the workflow as these technologies can acquire on-site data quickly and accurately (in most cases, <1h). After acquiring raw data, data interpretation follows up to extract useful information (e.g., discontinuity, water inflow and profile information) by using corresponding interpretation methods such as edge detection (Azarafza et al., 2019; Leu and Chang, 2005), clustering method (Chen et al., 2016), deep learning (Chen et al., 2021a; Chen et al., 2021b), which have proven to be intelligent approaches with small time consumption but high accuracy. Then, the extracted information is used for aggregation to give final revised support schemes. In this step, numerical analysis and artificial intelligence (AI)-support approaches are adopted for their superiority of providing results in a short period of time. Specifically, artificial neural network (ANN) (Nie et al., 2018), genetic algorithm (GA) (Alvarez-Fernandez et al., 2020; Liu et al., 2019), particle swarm optimization (PSO) (Li et al., 2021), expert system, and other AI-based approaches have proven to be effective in tunnel support design. Finally, revised support schemes are fed back to the site for construction.

It should be noted that though the proposed conceptual framework of JIT design of rock tunnel support includes different methods and conceptions, its main aim is to improve the efficiency of each step in the "acquisition-interpretation-aggregation" workflow. The mentioned methods and conceptions in Fig. 1 are summarized from previous studies and proven to be effective for tunnel support design. Some conventional methods such as manual sketch of the exposed tunnel face are not included in the proposed framework as they are neither time-saving nor accurate.

2.2. Comparison with dynamic design of rock tunnel support

Fig. 2 presents the comparison between the proposed JIT design of rock tunnel support and widely-known dynamic design of rock tunnel support. It can be seen that the focus of the dynamic design of rock tunnel support lies in the modification of original support schemes using newly-exposed data during construction. In this practice, some time-consuming data acquisition methods such as manual sketch of tunnel face, which may take as long as 4 h (Huang et al., 2019), and inaccurate data interpretation and aggregation methods such as empirical analysis are all included.

By comparison, the focus of JIT design of rock tunnel support lies in reducing time consumption and increasing accuracy in the data "acquisition-interpretation-aggregation" workflow. It does not mean that JIT design of rock tunnel support is completely different from the dynamic design of rock tunnel support. Actually, JIT design of rock tunnel support is still within the scope of dynamic design of rock tunnel support, but it raises higher requirements for dynamic design of rock tunnel support that the revision of original support schemes should be made in time when geological conditions are newly exposed during construction, while the latter one focuses on the revision of the preliminary design parameters using on-site data regardless of the speed or the efficiency.

Hence, in the workflow of JIT design of rock tunnel support, intelligent acquisition and interpretation methods are recommended such as

DP and TLS. Data- and knowledge-driven methods are also recommended for their competence of improving accuracy.

2.3. DT-enabled JIT design of rock tunnel support

DT and JIT design of rock tunnel support share one common goal, which is to improve efficiency. Hence, in this study, DT is incorporated into JIT design of rock tunnel support to further improve the performance of JIT design of rock tunnel support. The SERIC modeling method is proposed and integrated into JIT design of rock tunnel support to improve the design efficiency of rock tunnel and interconnect key technologies in JIT design of rock tunnel for better comprehension and application of DT in JIT design of rock tunnel, trying to provide a paradigm shift for JIT design of rock tunnel.

2.3.1. SERIC modeling of DT

After the concept of DT was introduced in a presentation to industry in University of Michigan, Prof.Grieves further expounded the definition of DT and stated that DT (DT Prototype and DT Instance) was operated in a DT Environment. On this basis, we subdivide DT into DT Scene (DTS), DT Entity (DTE), DT Relationship (DTR), DT Incident (DTI) and DT Controller (DTC) for better comprehension and application of DT in AEC industry.

DT scene is similar to the concept the DT environment proposed by Grieves and Vickers (2017). What really sets them apart is that one DT environment can incorporate many DT scenes. Here, DT scene refers to the specific application scene in AEC industry, and it is a specific description of the task and action within certain time and space in order to solve a practical engineering problem. Taking rock tunnel construction as an example, real-time acquisition of newly-exposed tunnel face data and subsequent rock classification based on the acquired data is a DT scene which aims to use DT technologies to assist tunnel design (Yang et al., 2021). In addition, intelligent control of jumbos on site is also a DT scene which aims to use DT technologies to assist tunnel construction (Navarro et al., 2018).

DT entity is the entities in the DT scene. According to the DT paradigm, i.e. physical twin, virtual twin and data flow, equipment, stuff, structures and other physical entities (PE) that have a counterpart in virtual space and have data flows between them can all be considered as DT entity. DT entity contains geometry, properties, states, forms, mechanisms, connections, and behaviors of entities in DT. Taking the lighting design inside the tunnel as an example (Shen et al., 2022), tunnel structure and luminaires are two DT entities as they both have counterparts in the virtual environment and data collected in the virtual environment are used to revise parameters of physical entities.

DT relationship describes relationships between DT entities, including information correlation, spatial correlation, physical correlation, aggregation correlation and so forth. Understanding how the DT entities are correlated is crucial as their relationships determine how the entire DT should be constructed. Specifically, spatial correlation, physical correlation and aggregation correlation determine how the model in virtual space is established, while information correlation and aggregation correlation determine how data are transmitted, aggregated and analyzed.

DT incident refers to events in a DT scene. Event-driven method uses real-time data collected by the production system to capture the disturbance events in the production process and explores the real-time changes of system performance under random disturbance scenario (Gou et al., 2021). Hence, in the modeling of DT, we use DT incident to denote events in the whole DT system which can trigger data exchange between digital- and physical-twin. For instance, in the study carried out by Ritto and Rochinha (2021), damage detection is one DT incident and triggers data exchange between physical twin, discrete computational model and stochastic computational model. In the study conducted by Yuan et al. (2016), structure monitoring is one DT incident and triggers data updates and remote control between virtual model and physical

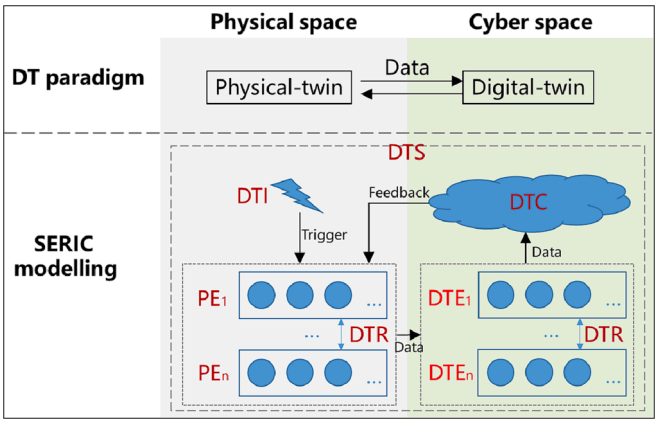


Fig. 3. Relationship between SERIC modeling and DT paradigm.

structure. It should be noted that one DT scene can exist different DT incidents as in some complex tasks there may be different data exchange aims.

Last but not least, DT controller is the data and information processing and analysis center in DT scene. DT controller is the core component in virtual space which incorporates algorithms, models, knowledge base and so forth. Data collected from the physical-twin are the input to DT controller, and orders and other feedbacks to the physical-twin are the output of DT controller. Up until fairly recently researchers have been allocating efforts to improve the performance of DT controller, such as introducing more computationally-power approaches (Chong et al., 2014) and investigating better-performing models (Liu et al., 2021).

The proposed SERIC modeling method is consistent with DT paradigm, as shown in Fig. 3. The relationships between DT scene, DT incident, DT entity, DT relationship and DT controller are also demonstrated in Fig. 3. In a specific DT scene, DT incident is the trigger that causes interaction between physical space and virtual space. Data collected from PEs in the physical space are transmitted to DT entities and DT controller in the virtual space for processing and analysis. It should be noted that DT entities in the virtual space are the digital-twin of the PEs in the physical twin. The results or orders given by the DT controller are then fed back to PEs in the physical space.

2.3.2. DT-enabled JIT design of rock tunnel support

The overall paradigm of DT in the JIT design of rock tunnel is presented in Fig. 4, where DT entity (and its counterpart PE in the physical space), DT relationship, DT incident and DT controller, as well as their mutual relationships, are depicted.

The DT scene of JIT design is clear, which is to use advanced data acquisition method on site, interpret raw data and give support parameters quickly to implement JIT design of rock tunnel.

The entities involved in the design and construction of rock tunnel can be divided into four categories: tunnel structure, geological body, people and mechanical equipment. Considering the specific requirements of JIT design of rock tunnel, tunnel structure, support structure, geological body, and mechanical equipment are four main entities to be modelled in the DT. Specifically, DT entity of tunnel structure (DTE_{tunnel}) and support structure ($DTE_{support}$) is modelled using parametric modeling method based on *Revit Dynamo*, DT entity of geological body (DTE_{geo}) is established using geological information system (GIS) technology, and DT entity of mechanical equipment ($DTE_{equipment}$) is also modelled in the building information modelling (BIM) environment.

DT relationship between the DT entities of JIT design of rock tunnel mainly contains spatial correlation, information correlation and aggregation correlation. To be specific, DTE_{tunnel} has spatial correlation with DTE_{geo}, spatial correlation and aggregation correlation with $DTE_{support}$,

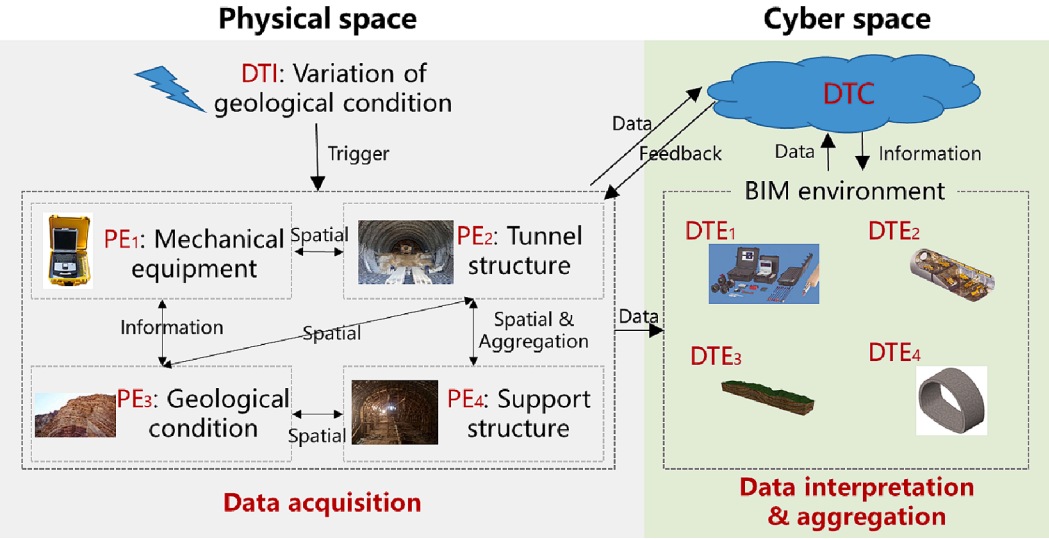


Fig. 4. Paradigm of DT in JIT design of rock tunnel.

and spatial correlation with $DTE_{equipment}$. DTE_{geo} has information correlation with $DTE_{equipment}$ and spatial correlation with $DTE_{support}$.

DT incident in the DT scene of JIT design of rock tunnel is mainly the variation of geological body condition. Once the geological body exposed during construction is different from that in the survey stage, PE of mechanical equipment (counterpart of $DTE_{equipment}$ in the physical space) such as TSP begins to work and acquires on-site data from PE of geological body. With the interpretation and aggregation of raw data, the updated support parameters are then given by DT controller using specific algorithms. At last, the analysis results guide the implementation of PE of support structure.

In the JIT design of rock tunnel, DT controller contains two parts: interpretation of raw data for useful information and aggregation of multi-source information to give support parameters. To be specific, in this study, interpretation of geological prospecting data using geostatistical method (see **Subsection 3.2**), empirical method to give support parameters using rock mass rating (RMR) value and BIM-based approach to aggregate all source of information(see **Subsection 3.3**) to improve construction management and design efficiency are all components of DT controller.

3. Framework implementation: Case study of Laoying tunnel

The proposed JIT design method of rock tunnel is applied and validated in Laoying Tunnel, which is a practical engineering project in Yunnan Province, China. As an extra-long tunnel with a length of more than 11 km, Laoying Tunnel passes through many different strata and the geological conditions are complex, which poses a challenge on the design of the tunnel. Based on the paradigm depicted in Fig. 1, this section tackles the JIT design of Laoying Tunnel. Location of Laoying Tunnel, as well as its profile, is presented in Fig. 5.

Implementation of JIT design of rock tunnel can be divided into three steps, i.e., acquisition of on-site raw data, interpretation of on-site raw data and aggregation of the interpreted results to obtain revision of preliminary design schemes. In Laoying Tunnel, tunnel seismic prediction (TSP) is selected as the data acquisition approach, based on which the -subsequent data interpretation and aggregation are conducted. Time consumption, accuracy and degree of automation are three main indicators influence the performance of JIT design of rock tunnel. Specifically, data acquisition using TSP and interpretation using geostatistical method can fulfill the requirements of time consumption, accuracy and degree of automation, as data acquisition time can be controlled within 30 min, geostatistical method can eliminate uncertainty in the data and improve prediction accuracy, and the whole interpretation and aggregation process can be automatic.

3.1. Data acquisition

TSP is a widespread non-destructive geophysical techniques of ahead geological forecast to detect water bodies, fissures and fractured rock ahead of the tunnel face (Bu et al., 2019). The principle of TSP geological forecast is to receive artificial seismic waves formed by low-dose explosives near the tunnel face(Lin and Li, 2013). The reflected waves are usually associated with discontinuities of rock masses. Due to its long detection distance and high prediction accuracy, TSP has become one of the most commonly-used methods for tunnel ahead geological forecast (Fu et al., 2018). As for the acquisition time, use of TSP on site to acquire data can be as quick as 30 min, which can meet the demands of rock tunnel JIT design.

In this study, the section from K12 + 100 to *K*11 + 020 was selected as the study area. A total of 9 TSP has been carried out in this area. A portion of the acquired data is shown in Table 1.

3.2. Data interpretation

Generally, data acquired by TSP include the wave type, *P* and *S* wave velocities (V_p and V_s), longitudinal and shear wave velocity ratio (V_p/V_s), and magnitude of the wave received. Once data of TSP are obtained, elastic property parameters of rock media can be calculated as the velocity of seismic wave in rock mass depends on the density, rock strength, groundwater condition, spacing, and filling state of discontinuities. Conventionally, TSP parameters are associated with the Poison's ratio (Eq. (1)) and Young modulus (Eq. (2)) to predict unfavorable geology before the tunnel face (Shi et al., 2014).

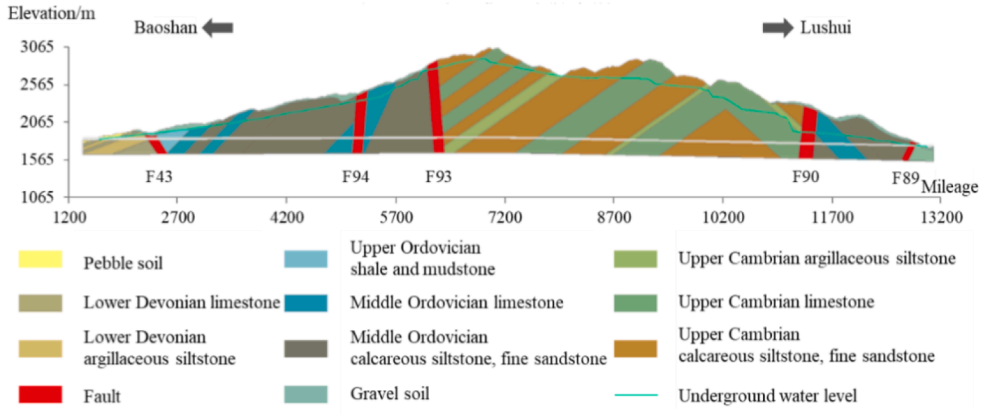
$$\mu = \frac{V_p^2 - 2V_s^2}{2\left(V_p^2 - V_s^2\right)} \tag{1}$$

$$E = \rho V_s^2 \left(\frac{3V_p^2 - 2V_s^2}{2V_p^2 - V_s^2}\right) \tag{2}$$

Where V_p is the longitudinal wave velocity, V_s is the shear wave velocity, and ρ is the rock density. The calculated mechanical parameters are then used for further numerical analysis such as analysis of joint rock mass stability using discrete unit analysis method (Fan et al., 2021). In addition, some researches also applied calculated mechanical



(a)



(b)

Fig. 5. (a) Location of Laoying Tunnel and (b) Tunnel profile.

Table 1
A portion of acquired TSP data and RMR value in the survey stage.

Mileage	RMR value	V_p(m/s)	V_p/V_s
K12 + 320	45	3051	1.79
K12 + 300	52	3209	2.13
K12 + 280	51	3533	1.92
K12 + 240	32	2872	1.62
K12 + 220	38	2979	1.63
K12 + 160	57	3451	1.97
K12 + 080	43	3156	1.73
*K*11 + 960	43	3058	1.77
*K*11 + 860	55	3994	1.92
*K*11 + 460	38	2560	1.23
*K*11 + 340	39	3056	1.57

parameters to rock classification such as rock mass rating (RMR) value (Bu et al., 2018; Shi et al., 2014). However, the numerical analysis can consume much time in the case when the model is complex, and the existing rock classification methods using TSP data are mostly qualitative without considering the quantitative relationship between TSP data and rock classification parameters. Moreover, due to the complex physical coupling mechanism of wave velocity in discontinuous media and the uncertainty of RMR rating parameters, it is difficult to accurately estimate RMR or obtain a universal relationship between wave velocity and RMR by using wave velocity data. Therefore, it is necessary to update the model in real time based on the engineering data on site. Schematic diagram describing the estimation of RMR value and updating of model using TSP data and tunnel face data is presented in Fig. 6. In time n + 1, the estimation of RMR value in time n + 2, n + 3, n + 4 and so forth will be updated and may be different from that in time n.

In this study, we propose a geostatistical method to accurately and

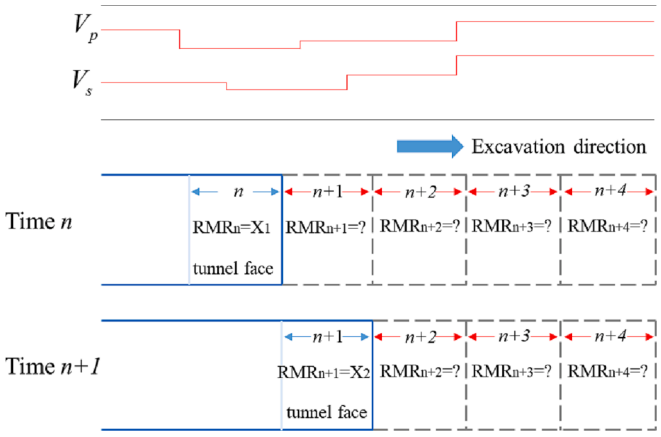


Fig. 6. Updating of RMR value during excavation.

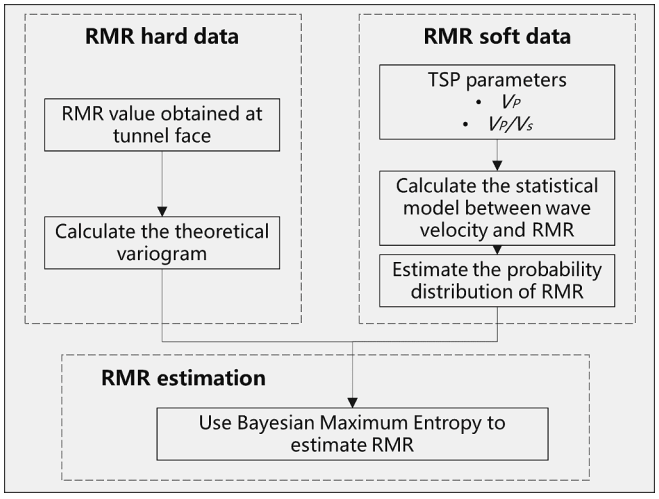


Fig. 7. Workflow of data interpretation of TSP data.

quickly interpret the TSP raw data and obtain the estimation of rock quality in front of the tunnel face. The overall methodology is presented in Fig. 7.

As shown in Fig. 7, we use RMR hard data to denote the RMR value based on data at the tunnel face, and RMR soft data to denote the RMR value to be predicted by TSP data in front of the tunnel face. Firstly, the variogram, which is defined as the expected value of a squared difference (Isaaks and Srivastava, 1989) (see Eq. (3)), is used to quantify the spatial variability of the RMR value. The range of hard data involved in the prediction of RMR value can be determined by fitting the theoretical variogram and calculating the variation of the variogram.

$$2\gamma(h) = Var[Z(x) - Z(x+h)] = E\{[Z(x) - Z(x+h)]^2\} \tag{3}$$

where Z is a stationary random function with known mean m and variance σ^2 , and for all locations $\times$ in the study area $m(x) = m$ and $\sigma^2(x) = \sigma^2$ as Z is independent of location.

Then, as for the soft data, a linear model is adopted to represent the relationship between wave velocity and RMR for the convenience of engineering use, as shown in Eq. (4).

$$y = ax + b + \varepsilon \tag{4}$$

where y denotes the RMR value, x is the wave velocity data matrix, a and b are the parameters to be updated, and ε describes the observation error.

To update the parameters in Eq. (4), it should be noted that data on

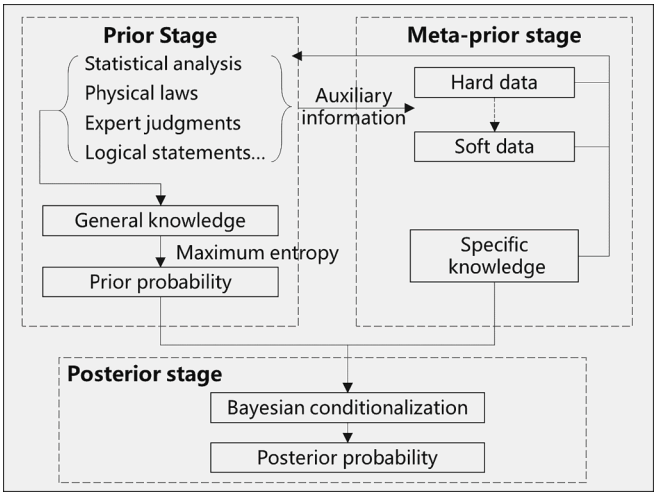


Fig. 8. Workflow of the BME method (()).
adopted from He and Kolovos, 2018

site are obtained gradually and round-by-round. In other words, parameters unknown at the previous round of tunneling may become known at the later round. Hence, the prediction of RMR value in front of the tunnel face should also conform to this characteristic so as to implement dynamic prediction. To this end, Bayesian sequential updating (see Eq. (5)) is adopted in this study to integrate prior knowledge (prediction in the previous round) and observation (data at the tunnel face) to update the model parameters, which has been widely used in geological prediction (Yang and Lam, 2018; Yao et al., 2020).

$$P([03B8]|x,y) = \frac{P(y|x,[03B8])p(\theta)}{\int P(y|x,[03B8])P(\theta)d\theta} = kP(y|x,[03B8])P(\theta) \tag{5}$$

where $\theta = [\mu_a, \mu_b, \sigma_a^2, \sigma_b^2]$ is the model parameters to be updated, y is the observation, x is TSP data, and k is the constant that normalizes the posterior probability of the parameter.

From Eq. (5) we know that $P(\theta)$ will be updated gradually with the construction going on, the implementation of which is that the posterior distribution of the parameters predicted in the previous step is the prior distribution of the parameters in the next step. Based on the Bayesian theory, the posterior distribution of parameters in the n-th round can be expressed as:

$$P(\theta_n) = P(\theta|data_n) = k_n P(data_n|\theta)P(\theta) \tag{6}$$

Correspondingly, the posterior distribution of parameters in the n + 1 round can be expressed as:

$$P(\theta_{n+1}) = P(\theta|data_{n+1}) = k_{n+1}P(data_{n+1}|\theta_n)P(\theta_n) \tag{7}$$

With the hard data and soft data, Bayesian maximum entropy (BME) is adopted to fuse data from different sources and with different precision to predict spatial variables. BME is one of the emerging methods in geostatistical predictions to overcome the shortcomings in classical geostatistical approaches as BME does not require too many constraints such as Gaussian distribution and linear estimations (Rezaei et al., 2020). Generally, the BME method is composed of three stages, namely the prior, *meta*-prior, and posterior stages, and the required knowledge in these stages can be classified into two groups: general knowledge and specificatory knowledge. At the first stage, the general knowledge is taken into account before any specificatory knowledge to maximize the system information entropy. Then, at the *meta*-prior stage, the available specificatory knowledge (including hard data and soft data) is organized in appropriate quantitative forms that can be explicitly incorporated into the BME formulation. Finally, at the posterior stage, the updated result incorporates both the general and specificatory knowledge (Modis

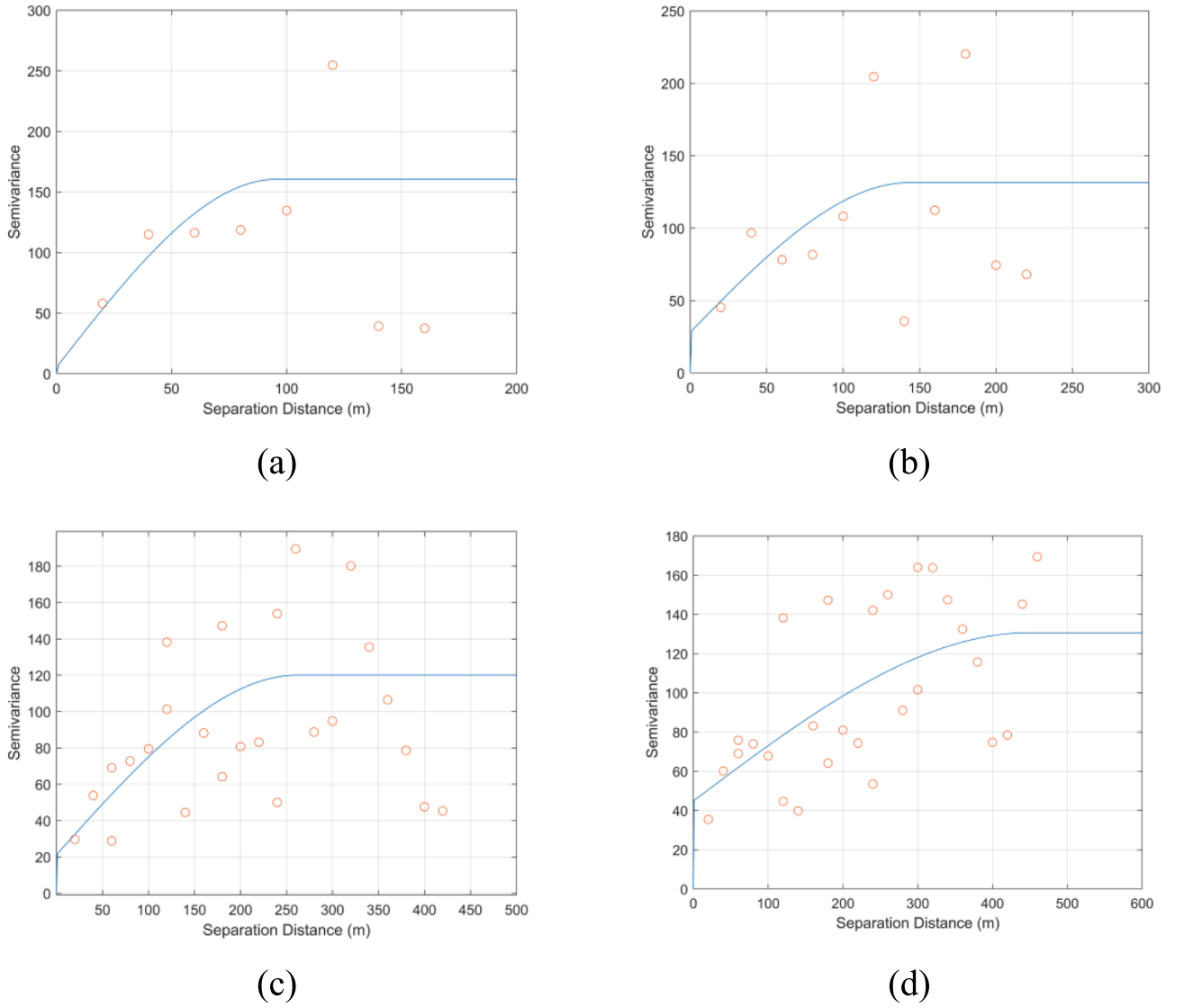


Fig. 9. Spatial variability of RMR value at four mileages: (a) $K11 + 900$; (b) $K11 + 760$; (c) $K11 + 480$; (d) $K11 + 360$ (The dots represent experimental variances calculated based on hard data at the four selected mileages, and the lines represent values fitted using spherical variogram).

Table 2
Parameters of the theoretical variogram at the four mileages.

Mileage	Nugget	Range	Sill
$K11 + 900$	5.27	95.54	155.17
$K11 + 760$	28.19	124.44	103.25
$K11 + 480$	21.17	261.93	98.88
$K11 + 360$	44.87	448.83	85.76

et al., 2013; Rezaei et al., 2020). Fig. 8 presents the schematic workflow for the BME method employed in this study.

3.2.1. Calculation of RMR hard data

In order to conform to the characteristics of dynamic acquisition of tunnel excavation information, four mileages, namely $K11 + 900$, $K11 + 760$, $K11 + 480$ and $K11 + 360$, were selected as the RMR prediction mileages, where the variogram was fitted with the measured RMR value after the tunnel face. That is to say, with the progress of tunnel construction, the hard data in the study area increased gradually. For the hard data at the four selected mileages, experimental variation of the data was calculated first and then spherical variogram was used to fit the experimental variation. Spatial variability of RMR value at the four mileages is presented in Fig. 9.

According to the theoretical variogram obtained in Fig. 9, the parameters calculated by the variogram at the four mileages are shown in Table 2.

From the fitting results, the range value increases with the increase of the measured data, which indicates that with the increase of the measured data for fitting variogram, the data range affecting the predicted mileage also increases, and more hard data will participate in the prediction process. At the same time, the increase of measured data leads to the gradual increase of the nugget, which shows that using a large amount of historical data for prediction cannot fully guarantee the improvement of prediction accuracy. In some areas, such as the fractured section of a fault, the prediction error caused by the internal randomness of surrounding rock parameters will be more obvious. Therefore, it is necessary to use the soft data formed by TSP to improve the prediction accuracy.

3.2.2. Calculation of RMR soft data

Based on the Bayesian sequential updating method described in **Subsection 3.2.1**, the parameter values of the RMR interpretation

Table 3
Parameters of the prediction model using V_p/V_s as independent variable.

Mileage	Parameter	Mean	Variance
$K11+900$	a	49.91	0.1554
	b	−42.56	3.2765
	σ_ε^2	0.2125	0.01254
$K11+760$	a	50.89	0.1846
	b	−19.32	1.8788
	σ_ε^2	0.4808	0.02986
$K11+480$	a	35.79	0.1264
	b	−18.54	0.7697
	σ_ε^2	0.3726	0.02964
$K11+360$	a	28.04	0.02854
	b	−4.97	0.8654
	σ_ε^2	0.6984	0.05498

model using wave velocity data can be obtained. With longitudinal and shear wave velocity ratio (V_p/V_s) as the independent variable, the parameters of the prediction model at the four specific mileages are shown in Table 3.

Taking mileage of $K11+360$ as an example, Fig. 10 presents the probability density function (PDF) of RMR interpretation values at the mileage of $K11+360$ under four interpretation models. It can be seen that with the gradual progress of tunnel excavation, the RMR soft data interpreted from wave velocity data are also gradually changing. Overall, as the measured data participating in the fitting gradually increase, the variance of the soft data gradually decreases, which indicates that the uncertainty of the soft data itself also gradually decreases with the increase of the measured data.

3.2.3. Prediction of RMR using BME method

By integrating hard data and soft data, the prediction of RMR value in front of the tunnel face at the four mileages is presented in Fig. 11. In addition, three evaluation indexes, i.e. maximum deviation, mean error and root mean square error, are used to assess the prediction results. The values of three evaluation indexes are shown in Table 4. As the construction progresses, the maximum deviation of predictions gradually decreases from 10.10 to 1.52 and the root mean square error gradually decreases from 5.11 to 0.93. It indicates that with the increase of hard data, adoption of V_p/V_s as the soft data to predict RMR can reduce the uncertainty of the prediction.

3.3. Data aggregation

With the updated RMR value using interpretation method in **Subsection 3.2**, the support parameters can be obtained correspondingly. As can be seen from Table 5, the one-to-one relationship between RMR value and support parameters can ensure the quick acquisition of support schemes.

In addition to the empirical approach to assess obtained RMR data, BIM is also adopted to aggregate multi-source heterogeneous data during construction such as design changes and construction progress, which are actually also crucial for accurate JIT design of rock tunnel. From one hand, BIM approach has undergone to a growing interest in tunnel engineering in the last few years for the evident benefit in terms of time and efficiency (Fabozzi et al., 2021), and from the other hand, BIM allows to facilitate the exchange and interoperability of information in digital format (Ryu et al., 2015), thus allowing stakeholders to rapidly and formally modify or evaluate design schemes. Hence, BIM approach is not only consistent with the aim of JIT design and DT, but also acts as an important part in DT paradigm as it facilitates data exchange and improve data interoperability.

Specifically, an automatic parametric modeling method of rock tunnel is proposed based on *Revit*, which can realize automatic updating of the model during construction using the engineering project progress and construction data, to provide guidance for JIT design and realize construction organization management. Then, a digital BIM platform (Zhu et al., 2018) is established to integrate multi-source heterogeneous data (such as TSP data, geological model, and survey data), display BIM model, and analyze the empirical results of support changes.

Based on the interpreted RMR value and the proposed paradigm of DT-enabled JIT design of rock tunnel, a BIM platform was established for Laoying Tunnel, where the DTE of geological conditions were depicted using GIS, and the establishment and updating of tunnel and support structure were implemented using parametric automatic modeling method. Spatial relationships between tunnel structure, support structure and geological model were established using DTRs mentioned in Fig. 4. Moreover, the BIM environment also integrated multi-source data according to the DTR of information aggregation. The support and tunnel model is presented in Fig. 12 **(a)**, and the BIM platform integrating GIS and multi-source data is presented in Fig. 12 **(b)**.

The application of such DT-enabled JIT design of Laoying Tunnel appears effective, and both design efficiency and accuracy can be guaranteed. Based on the BIM platform, several functions have been implemented: 1) automatic support determination using updated RMR value based on TSP data and empirical method; 2) automatic updating of tunnel support model for visualization and further analysis; 3) quick inquiry of geological conditions of tunnel excavation face using GIS data and other preliminary investigation data, which can assist the JIT design of tunnel support parameters; 4) visualization of construction progress and design schemes for end-users and stakeholders.

Compared with conventional design practice using manual sketch and empirical analysis, the proposed JIT design of Laoying Tunnel has several advantages. Firstly, the time has been greatly saved. According to experienced engineers on site, conventional design practice may need 2–3 days to complete the final design as manual sketch of tunnel face can take as long as 4 h, and the sketch results need to be sent to and

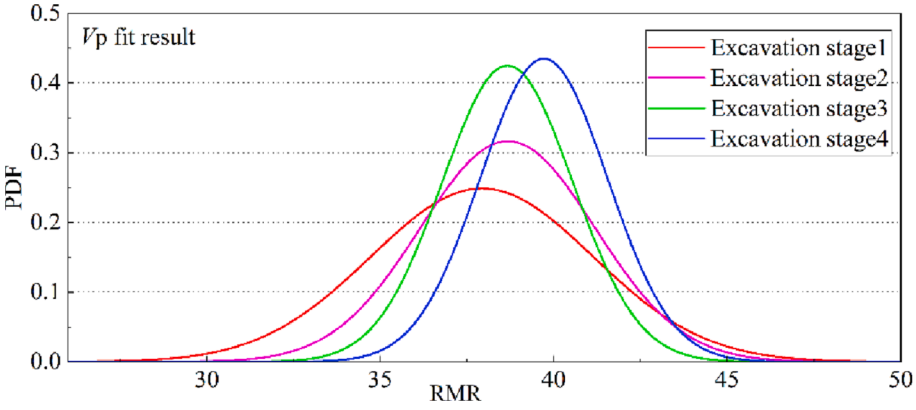


Fig. 10. PDF of RMR interpretation values at the mileage of $K11+360$.

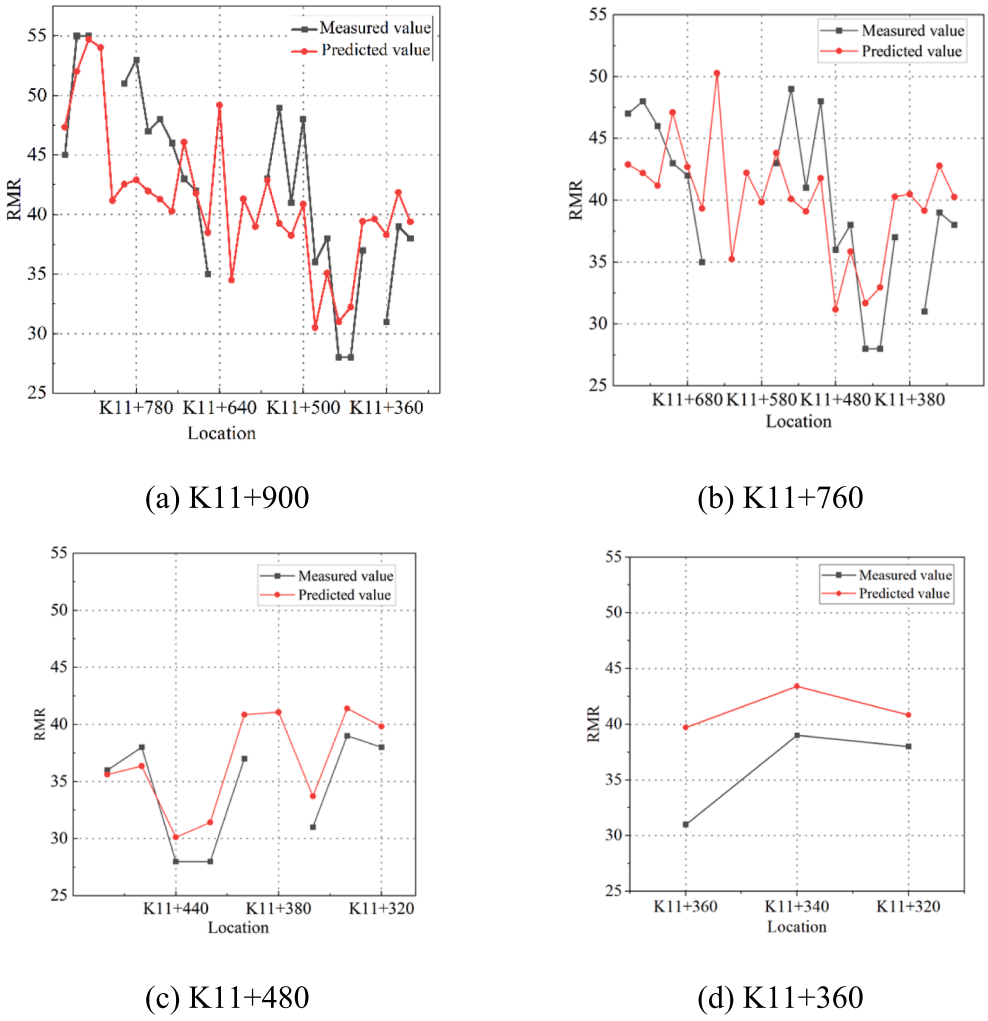


Fig. 11. Prediction of RMR value at four different mileages.

Table 4
Values of evaluation indexes at the four mileages.

	*K*11 + 900	*K*11 + 760	*K*11 + 480	*K*11 + 360
Maximum deviation	10.10	11.42	5.63	1.52
Mean error	−1.63	0.133	1.64	0.69
Root mean square error	5.11	5.36	3.13	0.93

interpreted by professional engineers for analysis. After all these have been done, the new design schemes need to be sent back to the site for evaluation and construction. Using TSP to acquire data, intelligent algorithms to interpret data, and BIM platform to aggregate and visualize results automatically, the whole JIT design process can be completed within two hours. Secondly, the accuracy of the results can be guaranteed, which partly owes to the use of BME method to accurately interpret TSP data. More importantly, data aggregation using BIM platform can allow different stakeholders to view and check the design results.

4. Limitations and future work

A conceptual framework of JIT design of rock tunnel and its implementation in a practical rock tunnel are presented in this paper, and results indicate that implementation of the proposed JIT design of rock tunnel outperforms conventional approaches in terms of time consumption and design accuracy. However, admittedly, this study bears several limitations that are noteworthy.

First of all, the proposed conceptual framework is based on previous studies on tunnel support design, and is validated using TSP data and intelligent algorithms to interpret and aggregate data. However,

Table 5
Recommended types of support based on RMR system (Rasouli, 2009).

RMR value	Anchoring Φ 20 mm	Shotcrete	Ribs
81–100	–	–	–
61–80	Locally bolts in crown, 3 m long, spaced 2.5 m, with occasional wire mesh	50 mm in crown where required	–
41–60	Systematic bolts 4 m long, spaced 1.5–2 m in crown and walls with wire mesh in crown	50–100 mm in crown, 30 mm in sides	–
21–40	Systematic bolts 4–5 m long, spaced 1–1.5 m in crown and walls with wire mesh	100–150 mm in crown, 100 mm in sides	Light ribs spaced 1.5 m where required
<20	Systematic bolts 5–6 m long, spaced 1–1.5 m in crown and walls with wire mesh. Bolt invert	150–200 mm in crown, 150 mm in sides and 50 mm in face	Medium to heavy ribs spaced 0.75 m with steel lagging and fore poling if required. Close invert

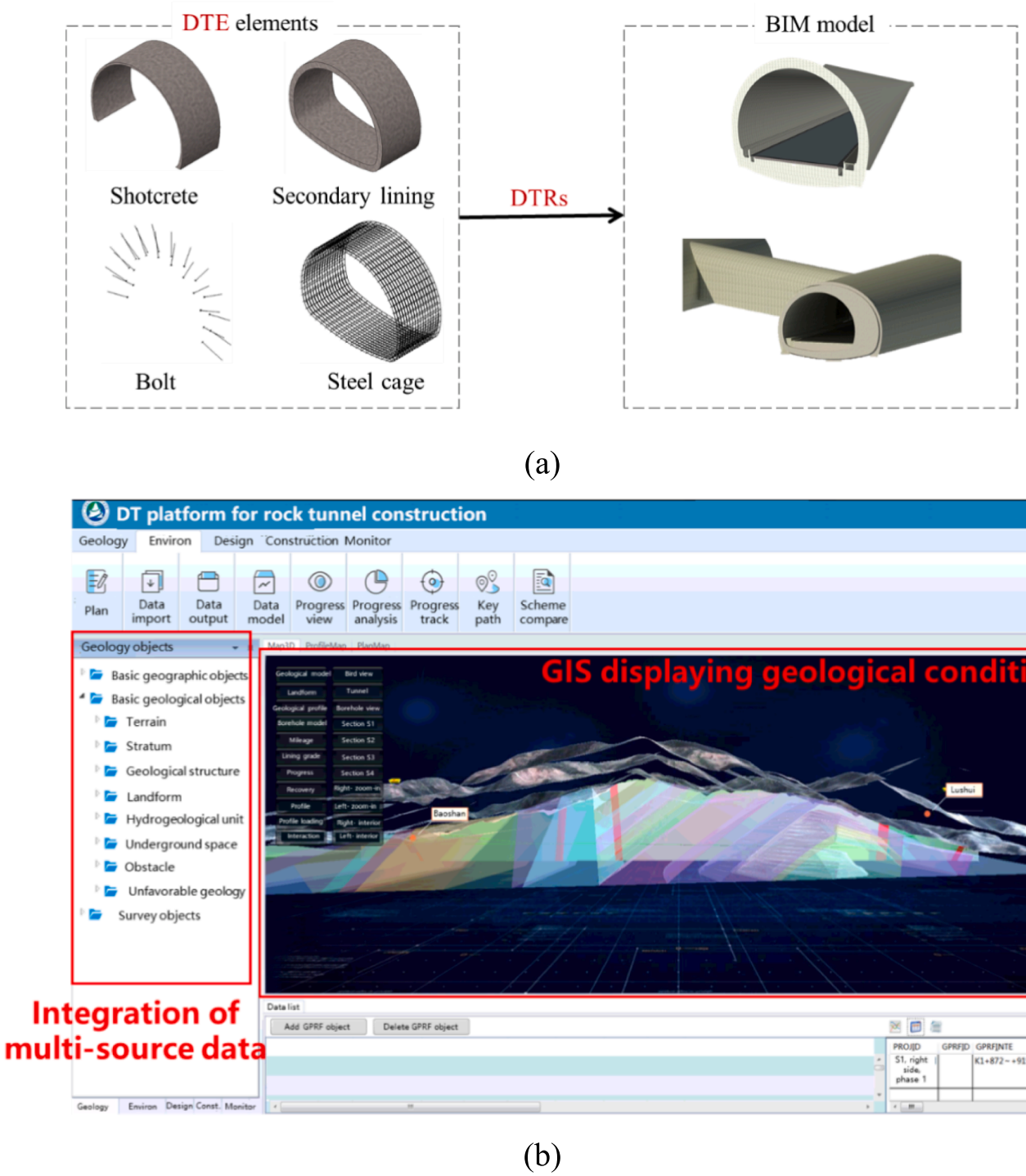


Fig. 12. BIM platform for data aggregation: (a) Tunnel and support model; (b) Geological conditions and integration of multi-source data.

currently in most cases, numerical analysis is widely used for support design. Though the proposed framework includes the numerical analysis (see Fig. 1), the case study did not adopt numerical analysis as the tool of data aggregation, which requires to be further studied in future researches.

Similarly, DP has become one major approach for data acquisition on site as its simplicity of obtaining data and possibility of interpreting data accurately. More importantly, using deep learning technologies to interpret images can be completed in seconds (Chen et al., 2021b). However, this study did not use this kind of acquisition approach. Hence, future research should take DP into account and use DP as one typical implementation of JIT design of rock tunnel.

In the proposed conceptual framework, the revised support schemes are fed back to the construction site for construction and evaluation. Hence, how to control jumbos and other intelligent construction machines to perform the construction task, and how to use on-site monitoring data to assess and evaluate the performance of JIT design, are two important tasks in the future.

5. Conclusion

This paper proposes a novel DT-enabled JIT design method for rock

tunnel. The main conclusions of the paper are listed below.

(1) a novel conceptual model, JIT design of rock tunnel, is proposed in this study to denote the intrinsic requirement of the rock tunnel support that support parameters need to be revised and determined when the rock mass are exposed during construction in a JIT way. Concepts and requirements of JIT design of rock tunnel are explained, as well as main indicators which influence the performance of JIT design of rock tunnel. Implementation of JIT design of rock tunnel can be subdivided into three steps, i.e., acquisition of on-site raw data, interpretation of on-site raw data and aggregation of the interpreted results to obtain revision of preliminary design schemes. A detailed workflow of the JIT design of rock tunnel is also presented.
(2) DT is incorporated into JIT design of rock tunnel support to further improve the performance of JIT design of rock tunnel support. The SERIC modeling method is proposed and integrated into JIT design of rock tunnel support to improve the design efficiency of rock tunnel and interconnect key technologies in JIT design of rock tunnel for better comprehension and application of DT in JIT design of rock tunnel.
(3) The proposed DT-enabled JIT design method of rock tunnel has been successfully implemented in a practical engineering project, where TSP is used as data acquisition method, and corresponding interpretation and aggregation have been conducted. Compared with conventional design practice using manual sketch and empirical analysis, the proposed JIT design of Laoying Tunnel outperforms in terms of time consumption (2–3 days vs 4 h) and accuracy (subjective vs objective). Moreover, the proposed method also facilitates information sharing between end-users and stakeholders.

CRediT authorship contribution statement

Xiaojun Li: Data curation, Writing – original draft, Writing – review & editing. **Li Tang:** Methodology, Writing – review & editing. **Jiaxin Ling:** Conceptualization, Methodology, Writing – review & editing. **Chao Chen:** Software, Validation. **Yi Shen:** Visualization, Investigation. **Hehua Zhu:** Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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